

ZIDANE VIRANI

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Education

Western University

Bachelor of Engineering in Software Engineering

Sep 2027

GPA: 3.8/4.0

- Courses: Computer Vision, Neural Networks, Deep Learning, Generative AI, Machine Learning, Statistical Learning, Probability & Statistics, Reinforcement Learning, Parallel Computing, Cloud Computing, Operating Systems, Data Structures & Algorithms
- Involvements: Director at Western AI, Academic Mentor for Freshmen, Outreach chair of Western Dental Association

Experience

Wisedocs (\$25M+ Series A Insurance AI startup): Incoming Machine Learning Engineer

Aug 2026 – Dec 2026

- Joining the Enterprise Insurance team, leading AI product development to forecast processing timelines across insurance claims

Bloomberg: Backend Engineer Intern

May 2026 – Aug 2026

- Built a REST API in Python serving pre-ranked bond screening results from a Redis cache and PostgreSQL, handling 160% more requests than the batch job it replaced across the full 300K+ bond universe
- Built a Python bond-screening pipeline that splits database reads across a thread pool with SQLAlchemy, loading the 300K+ bond universe in parallel to cut a run from 30 min to 2 min
- Added a Redis caching layer in Python for repeated issuer and pricing lookups across the screening run, cutting redundant database hits ~70% and shaving minutes off every full pass

Shopify: Applied Machine Learning Engineer Intern

Jan 2026 - Apr 2026

- Built a low-latency autocomplete model for Shopify Sidekick using Python, PyTorch, Hugging Face, and a fine-tuned Qwen-0.6B LLM, reducing merchant query time and deploying through Docker and Kubernetes to 5M+ active stores
- Improved LLM serving performance on H200 GPU clusters at Shopify through CUDA-level optimization of batching, tensor operations, and memory usage, increasing inference throughput 2.4x across high-volume merchant query workloads
- Built a Python and SQL training-data pipeline for Shopify Sidekick, using Polars and NumPy to clean 2M+ merchant-query logs and generate features, labels, and evaluation datasets for retraining workflows

Western University: Machine Learning Researcher

Jun 2025 – Sep 2025

- Built a computer vision pipeline in Python with PyTorch to detect debate interruptions, using ResNet-50 for image feature extraction and a Temporal Transformer for sequence modeling across 120+ hours of multimodal data, achieving 99.8% accuracy
- Built a Python pipeline using OpenCV and NumPy to decode 10M+ video frames, sync them with transcripts, and label which speaker was talking in each segment for model training
- Accelerated PyTorch video-sequence model training on 120+ hours of debate footage across H100 GPUs, reducing per-epoch runtime 45% through CUDA-level GPU programming in C++ and parallel data loading

Vector Institute: Software Engineer Intern

Apr 2025 - Jun 2025

- Built and deployed a Python backend for a voice-controlled assistant across 18 retirement facilities, containerizing speech APIs and background workers with Docker and Kubernetes to scale real-time request handling and save staff 1,200+ hours yearly
- Created backend data pipelines in Python and SQL to process 20K+ human-feedback signals, convert repeated user requests into structured training data, and improve assistant response quality by 38%
- Built a computer vision feedback module in Python/PyTorch using OpenCV and YOLOv8 to classify positive and negative user reactions from camera-based feedback, converting facial review signals into structured labels for assistant retraining

Projects & Awards

Founder & Lead Engineer, AI Content Engine ([Blog Link](#))

Jan 2023 – Dec 2024

- Sold a Python based Retrieval Augmented Generation (RAG) and recommendation pipeline for \$50,000 that used retrieval, GPT-4 generation, and engagement feedback loops to scale the platform to 120K+ monthly readers and \$2,800/month in net profit
- Owned deep learning models end-to-end, training a transformer (BERT) ranking model in PyTorch on 1M+ user engagement examples to improve recommendations almost 350% through iterative retraining

IBM Research ([Award Link](#)) | C++, Linux, Algorithms, Optimization

Jun 2026

- Earned global recognition on IBM Research's official website after solving an open algorithms challenge, by engineering a C++/Linux based competitive programming solution for an exponential optimization problem

Leadership

Aga Khan Development Network: Chair & Lead Engineer

Feb 2026 – Present

- Led 6 engineers and healthcare leaders to build an AI-powered full-stack fair-wage platform in India using a Python backend (FastAPI), React, OpenAI API and AWS, securing \$100K in funding and deploying to over 2,500 users

Technical Skills

- Languages & ML: Python, C++17/20, SQL, Go, Bash, PyTorch, CUDA, Hugging Face, ONNX Runtime, NumPy, pandas, Polars
- Systems & Tools: Linux, Docker, Kubernetes, AWS, Redis, Git, CMake, GDB, Google Benchmark